

USAMA IMRAN CHEEMA

Linux Systems Administrator & Cloud Infrastructure Engineer

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EXECUTIVE SUMMARY

Proactive **Linux Systems Administrator & Cloud Infrastructure Engineer** with **3+ years of intensive hands-on experience** in low-level Linux administration, POSIX systems engineering, server security hardening, and hybrid cloud operations (Azure & AWS). Proven track record of engineering high-throughput systems software in C/POSIX (mmap, atomic bitfields, lockless concurrency), developing automated security auditing toolchains, and maintaining 24/7 virtualized server clusters using **QEMU/KVM**. Skilled in deep networking diagnostics (TCP/IP, DNS, Nginx reverse proxy, firewalls) and 24/7 uptime monitoring. Dedicated to building, automating, and securing high-reliability production infrastructure.

CORE TECHNICAL SKILLS

Operating Systems & Virtualization:	Linux (Ubuntu Server, Debian, RHEL/Rocky, Arch Linux), QEMU/KVM, Libvirt, Proxmox, Docker, Kernel/POSIX APIs
Server Administration & Ops:	Systemd daemons, SSH Key Hardening, User/Group RBAC, Log Diagnostics (journalctl, dmesg), top, ss, df, cron
Cloud Platforms:	Microsoft Azure (Linux VMs, VNets, Subnets, NSGs, Blob Storage), AWS (EC2, VPC, S3, IAM, SigV4 Auth)
Networking & Security:	TCP/IP, DNS, DHCP, Nginx Reverse Proxy, SSL/TLS (Let's Encrypt), UFW/Iptables, Fail2ban, Threat Intelligence Auditing
Systems Scripting & Tools:	C / POSIX Systems APIs (posix_fallocate, pwrite, mmap), Python, Bash Automation, Go, Git/GitHub, Cloud-init

FEATURED SYSTEMS & INFRASTRUCTURE PROJECTS

down — High-Performance Concurrency & POSIX Transfer Engine

C, POSIX, Linux Kernel Primitives, AWS SigV4

- Engineered a lockless, multi-threaded systems utility in C/POSIX leveraging advanced Linux kernel primitives (posix_fallocate, atomic pwrite, and mmap).
- Implemented memory-mapped crash recovery utilizing an atomic bitfield tracking state per 512 KB chunk, resuming interrupted transfers without disk re-scanning.
- Designed an atomic work-stealing scheduler that dynamically bisects lagging tail-worker ranges to eliminate network bottlenecks.
- Integrated OpenSSL cryptographic verification (SHA-256, BLAKE2) and AWS SigV4 request authentication for cloud object storage transfers.

omasecurity — Automated Security Posture Auditor & Health Monitor

Python, Linux Security, Audit Tools

- Developed an automated Linux security auditing tool evaluating system hardening standards, service health, and software integrity.
- Implemented diagnostic checks for open network ports, unauthorized privilege escalation vectors, and misconfigured systemd units.
- Designed automated alerting routines to detect degraded service states and anomalous resource consumption across daemon processes.

omascan — Domain & Network Threat Intelligence Scanner

Python, Threat Intelligence APIs, DNS/SSL

- Built a network threat reconnaissance tool integrating multi-source threat intelligence APIs (VirusTotal, urlscan.io) for real-time security assessment.
- Automated network endpoint scanning, SSL certificate validation, and DNS record inspection to detect malicious infrastructure.

Enterprise Homelab & Multi-Node Linux Virtualization

QEMU/KVM, Bare-Metal Linux, Systemd, UFW, Nginx

- Architected a 24/7 self-hosted server environment powered by bare-metal Linux and QEMU/KVM virtual machines to simulate enterprise multi-tier environments.
- Configured local DNS, DHCP routing, static subnetting, and Nginx reverse proxies with automated SSL/TLS certificate renewal.
- Enforced defense-in-depth security: disabled root login, implemented ED25519 SSH keys, strict UFW firewalls, and Fail2ban intrusion protection.

EDUCATION & BACKGROUND

Self-Directed Computer Systems Curriculum

2022 – Present | OS Architecture, Kernel Internals, Networking, Cloud

Undergraduate University Studies

4 Semesters Completed | Analytical Reasoning & Scientific Research

CERTIFICATIONS (IN PROGRESS)

Red Hat Certified System Administrator (RHCSA EX200)

In Active Preparation – Live Lab Testing Track

Microsoft Certified: Azure Administrator (AZ-104)

In Active Preparation – Cloud Compute & VNets